

Unit-2 Software Project Scheduling [8 Hrs.]

2.1 Objectives of activity planning,

2. Work breakdown structure (WBS),

2.3 Network planning model:

 Critical path method (CPM),

 Program evaluation and review technique (PERT),

 Precedence diagramming method (PDM),

2.4 Shortening project duration,

2.5 Identifying critical activities.

2.6 Forward pass and Backward pass

GENERAL PRACTICE

- On large projects, lots of small tasks occur to accomplish a larger goal
 - Some tasks are dependent on other
 - Other tasks are independent
- Project manager's objectives
 - Define all project tasks
 - Build an activity network that depicts their interdependencies
 - Identify the tasks that are critical
 - Build a timeline depicting the plan and progress of each task
 - Track task progress to ensure that delay is recognized

WHAT IS PROJECT SCHEDULING?

- Scheduling distributes estimated effort across the planned project duration by allocating the effort to specific tasks
- During early stages of project planning, a macroscopic schedule is developed
- Later, each task is refined into a detailed schedule
- Project Scheduling can be viewed from two different perspectives
 - First view - an end-date for release of a computer-based system has already been established and fixed
 - The software organization is constrained to distribute effort within the prescribed time frame
 - Second view, assume that rough chronological bounds have been discussed but that the end- date is set by the software engineering organization
 - Effort is distributed to make best use of resources and an end-date is defined after careful analysis of the software
 - The first view is encountered far more often than the second



PROJECT SCHEDULING

2.1 Objectives of Activity Planning

Activity planning is a fundamental step in the project management process, particularly in the context of software project scheduling. It involves breaking down the project into smaller, manageable tasks or activities and assigning resources and timeframes to each one. The objectives of activity planning are as follows:

1. Task Clarity and Definition:

Activity planning ensures that each task or activity within the project is clearly defined. This includes specifying what needs to be done, who is responsible, what resources are required, and any specific criteria for success. Clarity in task definition minimizes ambiguity and misunderstandings, leading to more effective execution.

2. Sequencing and Dependencies:

Activity planning helps establish the sequence of tasks and identifies dependencies between activities. Understanding the order in which activities should be performed and their interrelationships is essential for efficient project execution.

3. Resource Allocation:

Effective activity planning allocates resources such as human resources, equipment, and budget to each activity. This ensures that the necessary resources are available when needed, reducing delays and resource conflicts.

4. Time Estimation:

Activity planning involves estimating the time required for each task. Accurate time estimates help in creating a realistic project schedule, which is a critical aspect of project management.

5. Risk Identification:

During the activity planning process, potential risks can be identified. This includes risks related to resource availability, dependencies, and task complexity. Identifying these risks early allows for proactive risk management.

6. Progress Monitoring:

Activity planning provides a basis for monitoring and tracking the progress of the project. By having a clear plan, project managers can compare actual progress against the planned schedule, making it easier to identify delays or issues.

7. Resource Optimization:

Activity planning enables the efficient use of resources. By allocating resources based on project needs, it helps prevent overallocation or underutilization of resources, which can impact project efficiency.

8. Stakeholder Communication:

A well-defined activity plan serves as a communication tool for stakeholders. It helps in conveying project details, timelines, and resource requirements to team members, clients, and other relevant parties.

9. Scope Management:

Activity planning contributes to effective scope management. It ensures that the project remains focused on its defined objectives and that any changes or additions to the scope are carefully considered and incorporated as needed.

In summary, the objectives of activity planning are to provide a structured and organized approach to project management. It leads to better control, reduced risks, and increased chances of project success by ensuring that tasks are well-defined, resources are allocated efficiently, and the project stays on schedule.

2.2 Work Breakdown Structure (WBS)

The Work Breakdown Structure, often abbreviated as WBS, is a foundational tool in project management, including software project scheduling. It is a hierarchical decomposition of the total scope of work to be carried out by the project team. The primary purpose of a WBS is to break down a project into smaller, more manageable components, making it easier to plan, execute, and control. Here are the key elements of a WBS:

1. Hierarchy:

The WBS is organized in a hierarchical manner, with the project's main objectives at the top level and progressively finer levels of detail as you move down the structure. This hierarchical structure reflects the natural decomposition of the project.

2. Decomposition:

Each higher-level element in the WBS is broken down into lower-level, more specific tasks or work packages. This decomposition continues until the project is broken down into its smallest manageable components.

3. Deliverables-Oriented:

The WBS focuses on deliverables, which are tangible or intangible outcomes of the project. These deliverables represent what the project is meant to achieve.

4. Control and Management:

The WBS provides a framework for project control and management. It helps project managers, team members, and stakeholders understand the project's scope and the individual tasks that need to be completed.

5. Scope Clarification:

By detailing the project's scope in the WBS, it ensures that everyone involved in the project has a clear understanding of what is included in the project and what is not.

Work Breakdown Structure (WBS) cont.

6. Responsibility Assignment:

The WBS assigns responsibilities for each task or work package to specific team members or groups. This clarifies who is accountable for what.

7. Estimation and Scheduling:

The WBS serves as a foundation for estimating the time, effort, and resources required for each task. This information is critical for creating a project schedule.

8. Change Management:

When changes are proposed or occur during a project, the WBS helps assess their impact. It allows for controlled changes without jeopardizing the project's integrity.

9. Communication:

The WBS acts as a communication tool. It helps convey the project's structure and scope to team members, clients, and stakeholders, promoting a shared understanding of project objectives.

10. Project Monitoring and Reporting:

The WBS facilitates project monitoring and reporting by providing a structured framework for tracking progress and reporting on completed tasks.

In summary, the Work Breakdown Structure is a vital tool in project management, enabling project teams to divide complex projects into manageable components, establish accountability, and facilitate clear communication. It serves as the foundation for project planning, scheduling, and control, making it an indispensable resource in the successful management of software projects.

2.3 Network Planning Models

Network planning models are essential tools in project scheduling, helping project managers and teams plan, execute, and monitor project activities efficiently. Three common network planning models are:

1. Critical Path Method (CPM):

- CPM is a deterministic model used to determine the longest path through a project's activities, which represents the minimum time needed to complete the project.
- Key Features:
 - Focuses on activity durations and dependencies.
 - Identifies the critical path, a sequence of activities that, if delayed, will delay the project.
 - Useful for projects with well-defined, known activity durations.

Network Planning Models

Network planning models are essential tools in project scheduling, helping project managers and teams plan, execute, and monitor project activities efficiently. Three common network planning models are:

2. Program Evaluation and Review Technique (PERT):

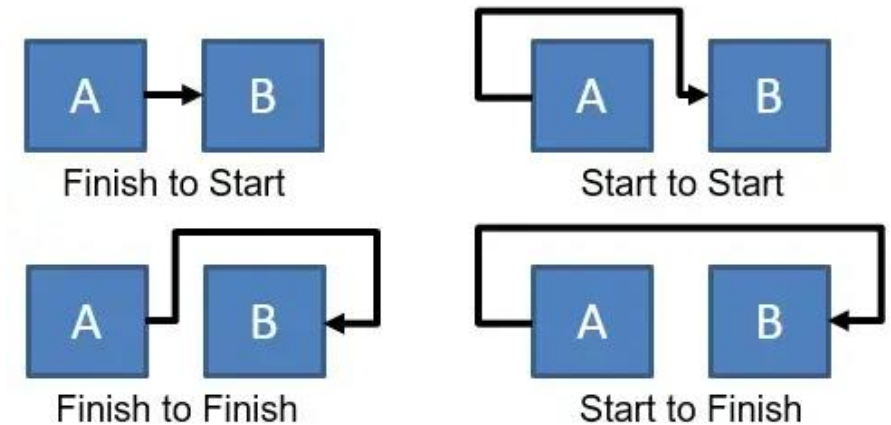
- PERT is a probabilistic model that incorporates uncertainty in activity durations, making it suitable for projects with significant variability.
- Key Features:
 - Uses three time estimates for each activity: optimistic, most likely, and pessimistic.
 - Calculates expected activity durations and project completion time using a weighted average.
 - Useful for complex projects with uncertain activity durations and high risk.

Network Planning Models

Network planning models are essential tools in project scheduling, helping project managers and teams plan, execute, and monitor project activities efficiently. Three common network planning models are:

3. Precedence Diagramming Method (PDM):

- PDM is a model that represents project activities and their dependencies using nodes and arrows, making it easy to visualize and analyze project schedules.
- Key Features:
 - Utilizes different types of dependencies, including finish-to-start, start-to-start, finish-to-finish, and start-to-finish.
 - Offers flexibility in representing complex relationships between activities.
 - Commonly used in software project management due to its visual clarity.



There are four ways to illustrate a project schedule network diagram. These form the connection between activities in a precedence diagram. They are:

- **Finish-to-Start (FS):** In this case, task B cannot start until task A has been completed
- **Start-to-Start (SS):** In this case, task B cannot start until task A starts
- **Finish-to-Finish (FF):** In this case, task A cannot be completed until task B is completed
- **Start-to-Finish (SF):** In this case, task B cannot be completed until task A starts

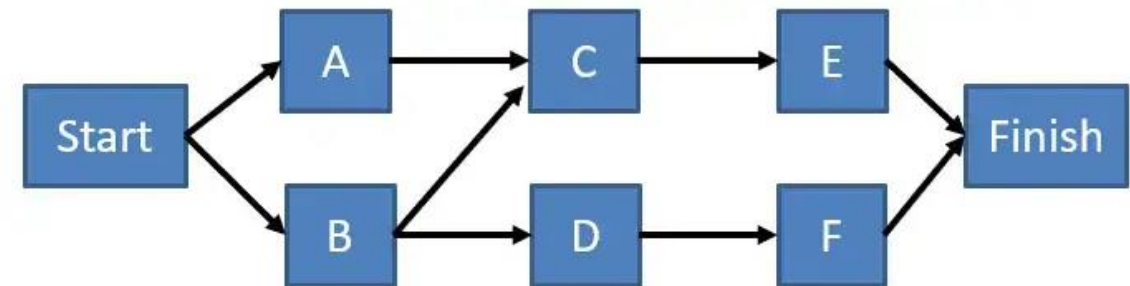
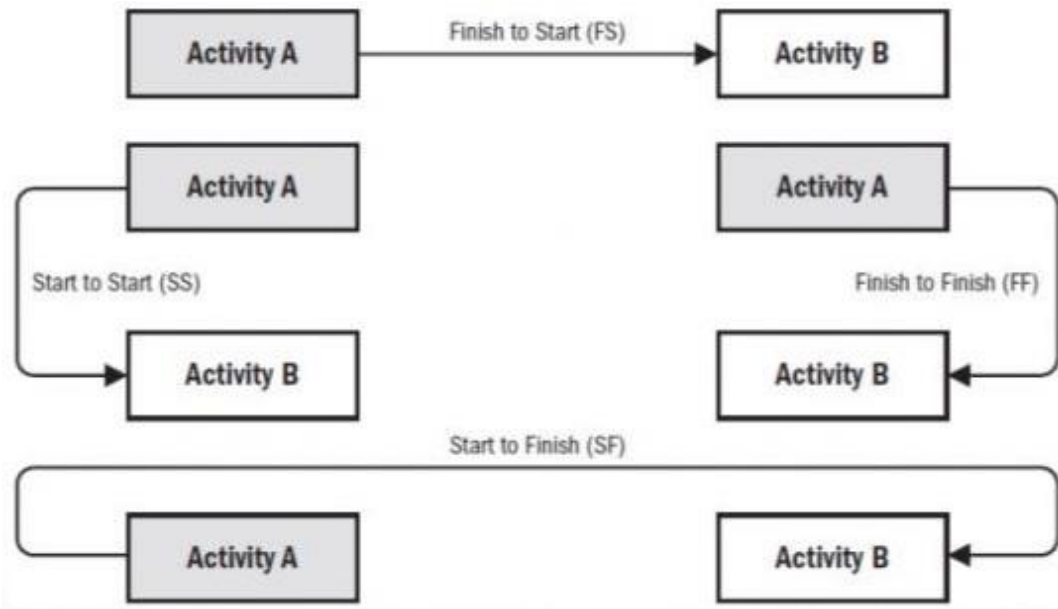


FIG: Precedence Diagram Method (PDM)

The Precedence Diagram is depicted by a chart with nodes and their relationships. An arrow connects two nodes to represent an active relationship. It's also called a nodal diagram or **network diagram**.

Network Planning Models

Network planning models are essential tools in project scheduling, helping project managers and teams plan, execute, and monitor project activities efficiently.

These network planning models provide project managers with tools to create project schedules, identify critical activities, allocate resources, and monitor progress. The choice of which model to use depends on the project's characteristics, such as the level of uncertainty in activity durations and the complexity of activity relationships. Each model offers unique advantages and can be tailored to the specific needs of a software project.

2 most common Network Planning Models

Both CPM (Critical Path Method) **and PERT** (Program Evaluation and Review Technique) **provide the user with project management tools to plan, monitor, and update their project as it progresses.**

Similarities between PERT and CPM

- Both follow the same steps and use network diagrams
- Both are used to plan the scheduling of individual activities that make up a project
- They can be used to determine the earliest/latest start and finish times for each activity

Differences between PERT and CPM

- PERT is probabilistic whereas CPM is deterministic
- In CPM, estimates of activity duration are based on historical data
- In PERT, estimates are uncertain and we talk of ranges of duration and the probability that an activity duration will fall into that range
- In routine projects where estimated time for each activity is known, CPM is a better tool to control both time and cost.

2.4 Shortening Project Duration

Shortening project duration, also known as project compression or schedule compression, is a crucial strategy in project management, especially for software projects. It involves finding ways to complete a project in less time without compromising its quality or objectives. Here are key considerations and techniques for shortening project duration:

1. Identify Critical Activities:

To shorten a project, you must first identify the critical activities. These are the tasks that, if delayed, will extend the project's overall duration. Critical path analysis, such as the Critical Path Method (CPM), is a valuable tool for this.

2. Fast Tracking:

Fast tracking involves performing critical path activities in parallel rather than sequentially. This approach can reduce project duration but requires careful management to avoid potential risks and conflicts.

3. Crashing:

Crashing involves allocating additional resources to critical path activities to complete them more quickly. It may increase costs but can shorten the project timeline significantly.

4. Resource Reallocation:

Review the allocation of resources throughout the project and consider redistributing resources to focus on critical path activities.

5. Scope Reduction:

Evaluate whether specific project features or requirements can be deferred to later phases or omitted to expedite project completion. Be cautious not to compromise essential functionality.

Shortening Project Duration (Cont.)

6. Risk Analysis:

Reassess project risks and create contingency plans to address potential delays. Proactive risk management can prevent time-consuming setbacks.

7. Parallel Workstreams:

For complex software projects, consider dividing the work into parallel workstreams, where teams work on different components simultaneously, reducing the overall project timeline.

8. Agile Methodologies:

Agile approaches, such as Scrum, promote iterative development, which can lead to quicker deliverables and more flexibility in accommodating changes.

9. Team Collaboration and Motivation:

A motivated and collaborative project team can often work more efficiently and resolve issues promptly, contributing to project schedule compression.

10. Monitoring and Control:

Continuously monitor project progress and adjust the plan as necessary. Effective control mechanisms can help identify delays early, allowing for timely corrective actions.

In summary, shortening project duration is an essential strategy to meet tight deadlines or adapt to changing project needs. It requires a combination of effective project management techniques, resource optimization, and risk mitigation. However, it's crucial to strike a balance between schedule compression and maintaining project quality and integrity. Careful planning and ongoing monitoring are key to successful project duration reduction.

2.5 Identifying Critical Activities

Identifying critical activities is a fundamental step in project management, especially when working with scheduling techniques like the Critical Path Method (CPM). Critical activities are those that, if delayed, will impact the project's overall duration. Here's how you can identify them:

1. Calculate Early Start (ES) and Early Finish (EF) times:

Begin by calculating the Early Start (ES) and Early Finish (EF) times for each activity. ES is the earliest point in time when an activity can start, and EF is the earliest time it can finish.

2. Calculate Late Start (LS) and Late Finish (LF) times:

Determine the Late Start (LS) and Late Finish (LF) times for each activity. LS is the latest point in time when an activity can start without delaying the project, and LF is the latest time it can finish.

3. Calculate Total Float (TF):

Total Float (TF) is the amount of time an activity can be delayed without affecting the project's overall duration. TF is calculated as $TF = LS - ES$ or $TF = LF - EF$ for each activity.

4. Identify Activities with Zero Total Float:

Activities with a Total Float of zero are critical activities. These activities have no flexibility in terms of time and must be completed on schedule to prevent project delays.

2.6 Forward Pass and Backward Pass

Forward pass and backward pass are techniques used in the Critical Path Method (CPM) to identify the critical path and calculate activity start and finish times. Here's how they work:

Forward Pass:

- The forward pass is the initial phase of CPM, used to calculate the Early Start (ES) and Early Finish (EF) times for each activity.
- The process begins at the project's start and progresses through each activity, determining when it can start and when it will finish based on the given durations and activity dependencies.
- The formula used for the forward pass is: $ES \text{ of the current activity} = \text{Maximum EF of its predecessors}$.
- The last activity's EF becomes the project's estimated duration.

Backward Pass:

- The backward pass is used to calculate the Late Start (LS) and Late Finish (LF) times for each activity, working backward from the project's end.
- It identifies the latest time an activity can start and finish without delaying the project's overall completion.
- The formula used for the backward pass is: $LF \text{ of the current activity} = \text{Minimum LS of its successors}$. The LF of the last activity becomes the project's duration.

Determining the Critical Path:

- After the forward and backward passes, you can identify the critical path by identifying activities with zero Total Float (TF). The critical path is the longest path of connected activities with zero TF, and it represents the minimum project duration.
- In summary, the forward pass calculates early start and finish times, while the backward pass determines late start and finish times. Activities with zero Total Float are critical, and the critical path is the sequence of these activities. These techniques are essential for effectively managing project schedules and identifying areas where time compression may be necessary.

Agenda

- ▢ Objectives of activity planning
- ▢ Work breakdown structure
- ▢ Bar chart
- ▢ Network planning model:
 - ▢ Critical path method (CPM)
 - ▢ Program evaluation and review technique (PERT)
 - ▢ Precedence diagramming method (PDM)
- ▢ Shortening project duration
- ▢ Identifying critical activities.

Network Planning Model

These project scheduling techniques model the project's activities and their relationships as a network.

- In the network, time flows from left to right.
- There are two best techniques:
 - CPM(Critical Path Method)
 - PERT(Program Evaluation Review Technique)
- Both use the “**Activity on Arrow**” approach to visualize the project as a network.
 - Activities are drawn as arrows joining circles, or nodes which represent the possible start or completion of an activity or set of activities.
- More recently a variation on these techniques, called **precedence networks**, has become popular. This method uses **activity-on-node** networks where activities are represented as nodes and the links between nodes represent precedence (or sequencing) requirements.

Network Planning Model

Formulating a Network Model

- The **first stage** in creating a network model is to **represent the activities and their interrelationships as a graph.**
- In activity-on-node we do this by representing activities as nodes (boxes) in the graph – the lines between nodes represent dependencies.

Rules for constructing networks:

- *A project network should have only one start node.*
- *A project network should have only one end node.*
- *A node has duration.*
- *Links normally have no duration.*
- *Precedents are the immediate preceding activities.*
- *Time moves from left to right.*
- *A network may not contain loops.*
- *A network should not contain dangles.*

Network Planning Model

- Formulating a Network Model
 - Labelling conventions:

Earliest start	Duration	Earliest finish
Activity label, activity description		
Latest start	Float	Latest finish

Earliest Start + Earliest Start will be calculated from Forward pass

Latest Start + Latest Start will be calculated from Backward pass

$\text{Float} = \text{Latest Start} - \text{Earliest Start}$

Critical path = Path that has zero float

Network Planning Model

Adding the time dimension

- Having created the logical network model indicating what needs to be done and the interrelationships between those activities, we are now ready to start thinking about when each activity should be undertaken.
- The critical path approach is concerned with two primary objectives:
 - planning the project in such a way that it is completed as quickly as possible; and
 - identifying those activities where a delay in their execution is likely to affect the overall end date of the project or later activities' start dates.
- The method requires that for each activity we have an estimate of its duration.
 - The network is then analysed by carrying out a **forward pass**, to calculate the earliest dates at which activities may commence and the project be completed, and a **backward pass**, to calculate the latest start dates for activities and the **critical path**.

Network Planning Model

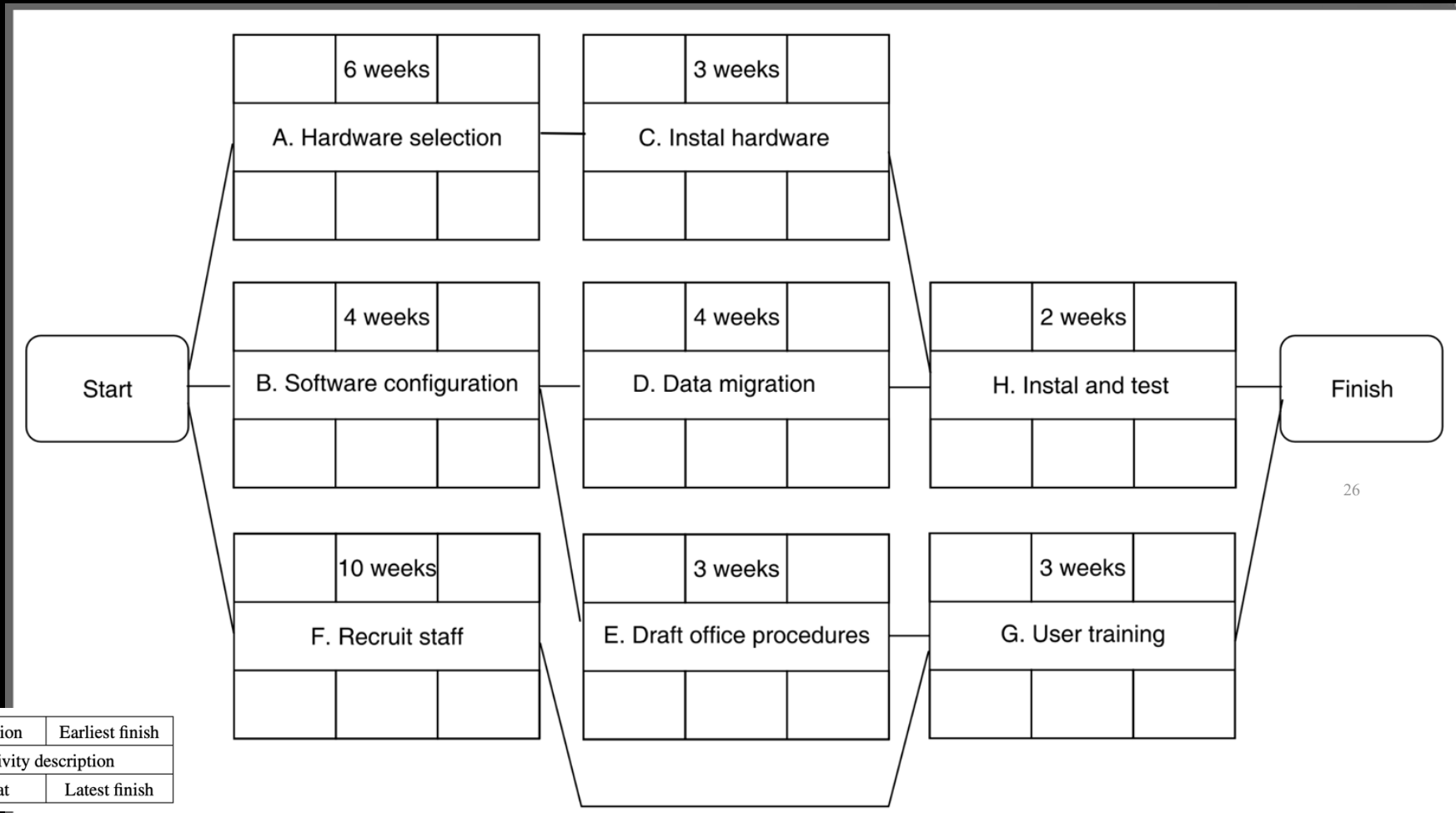
- The forward pass is carried out to calculate the earliest dates on which each activity may be started and completed.

An example project specification with estimated activity durations and precedence requirements

Activity		Duration (weeks)	Precedents
A	Hardware selection	6	
B	System configuration	4	
C	Instal hardware	3	A
D	Data migration	4	B
E	Draft office procedures	3	B
F	Recruit staff	10	
G	User training	3	E, F
H	Instal and test system	2	C, D

Network Planning Model

□ The precedence network would look like:

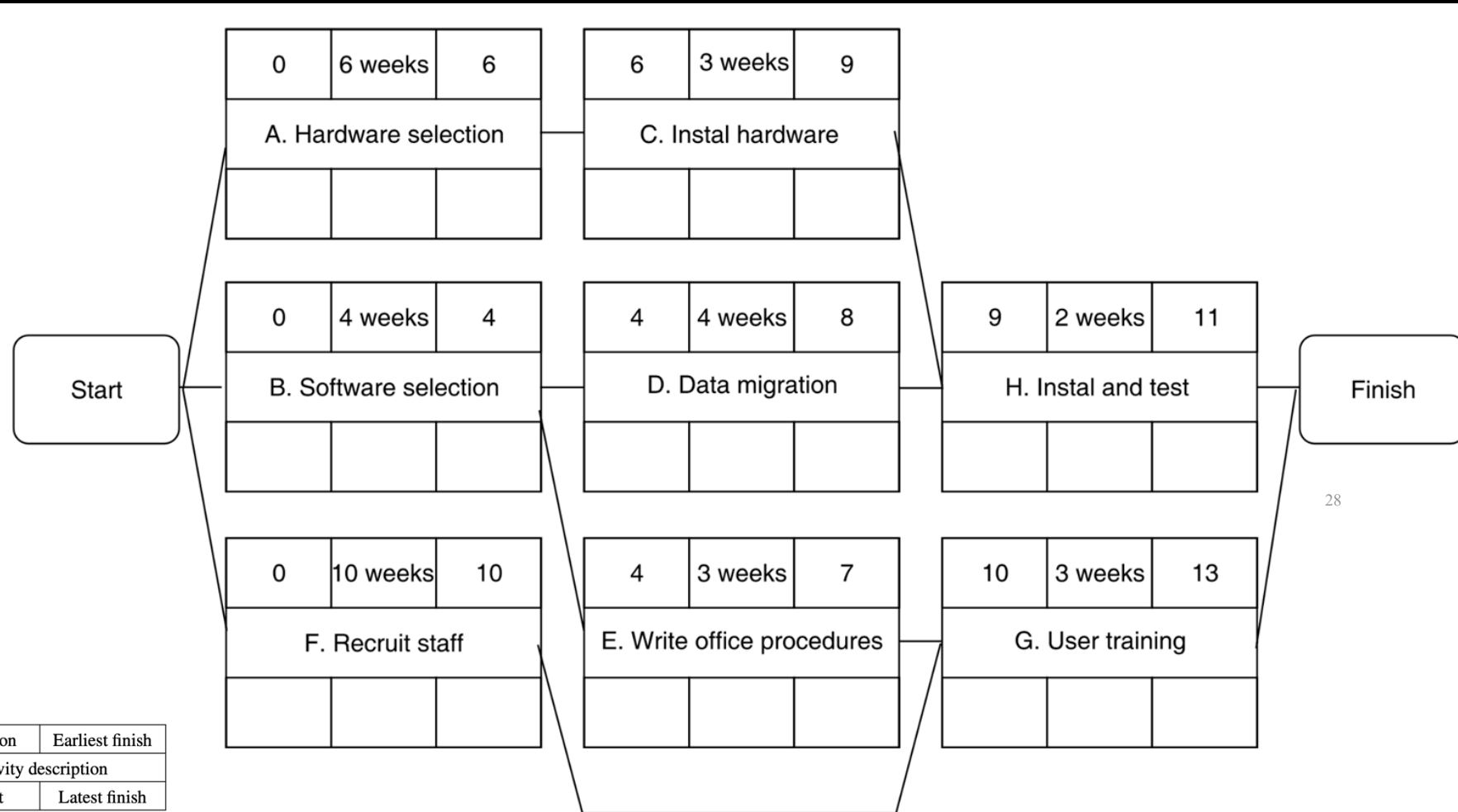


Network Planning Model (CPM - The Forward Pass)

- During the forward pass, earliest dates are recorded as they are calculated.
- **The forward pass rule:**
 - The earliest start date for an activity is the earliest finish date for the preceding activity.
 - Where there is more than one immediately preceding activity we take the *latest* of the *earliest finish dates* for those activities.

Network Planning Model (CPM - The Forward Pass)

□ The results of forward pass are shown below:



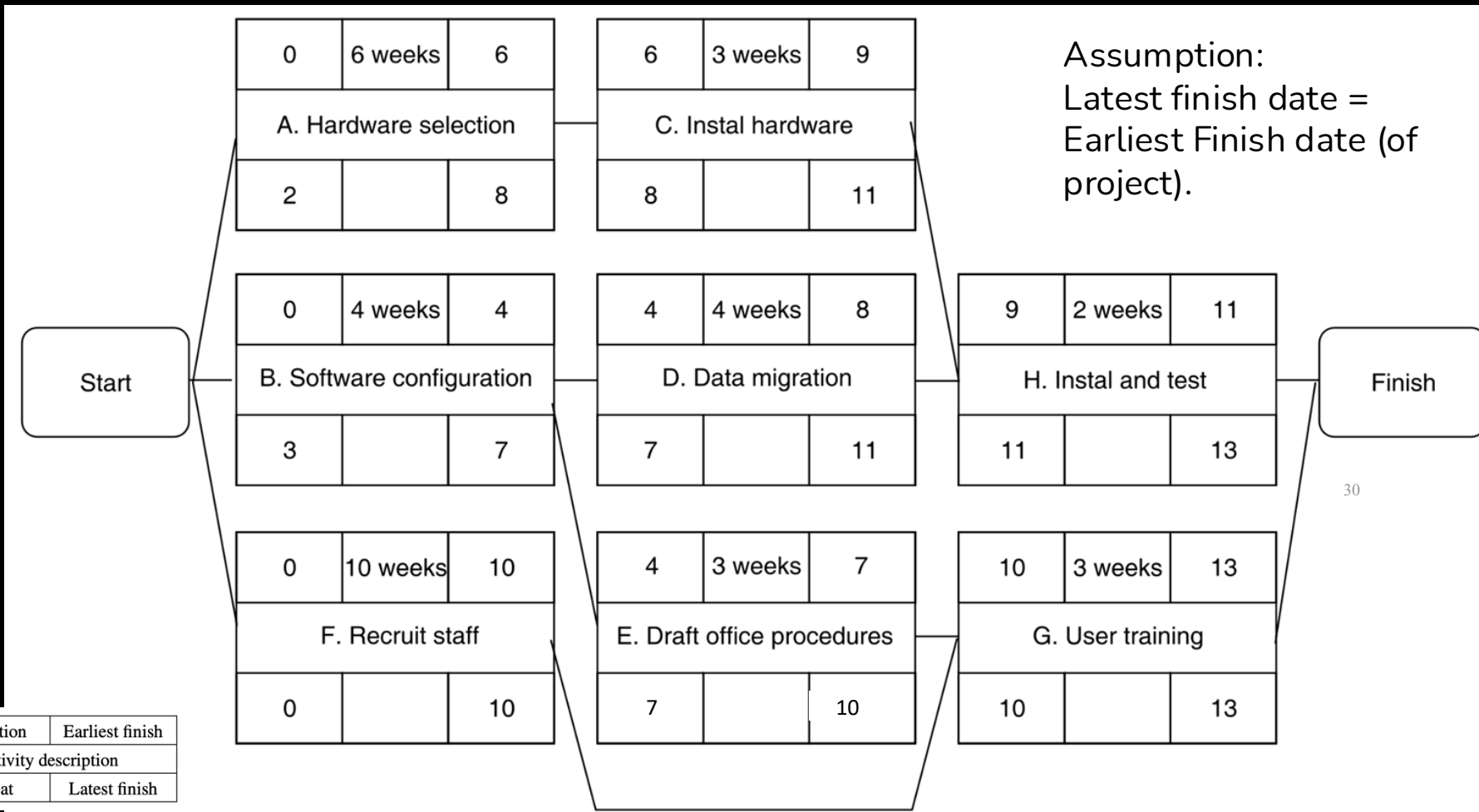
The forward pass is carried out to calculate the earliest dates on which each activity may be started and completed.

Network Planning Model (CPM - The Backward Pass)

- The second stage in the analysis of a critical path network is to carry out a *backward pass to calculate the latest date at which each activity may be started and finished without delaying the end date of the project.*
- In calculating the latest dates, we assume that the latest finish date for the project is the same as the earliest finish date – that is, we wish to complete the project as early as possible.
- **The backward pass rule:** The latest finish date for an activity is the latest start date for the activity that commences immediately that activity is complete. Where more than one activity can commence we take the earliest of the latest start dates for those activities.

Network Planning Model (CPM - The Backward Pass)

□ The results of backward pass are shown below:



The backward pass is carried out to calculate the latest dates on which each activity may be started and finished without delaying the end date of the project.

Earliest start	Duration	Earliest finish
Activity label, activity description		
Latest start	Float	Latest finish

Network Planning Model (CPM)

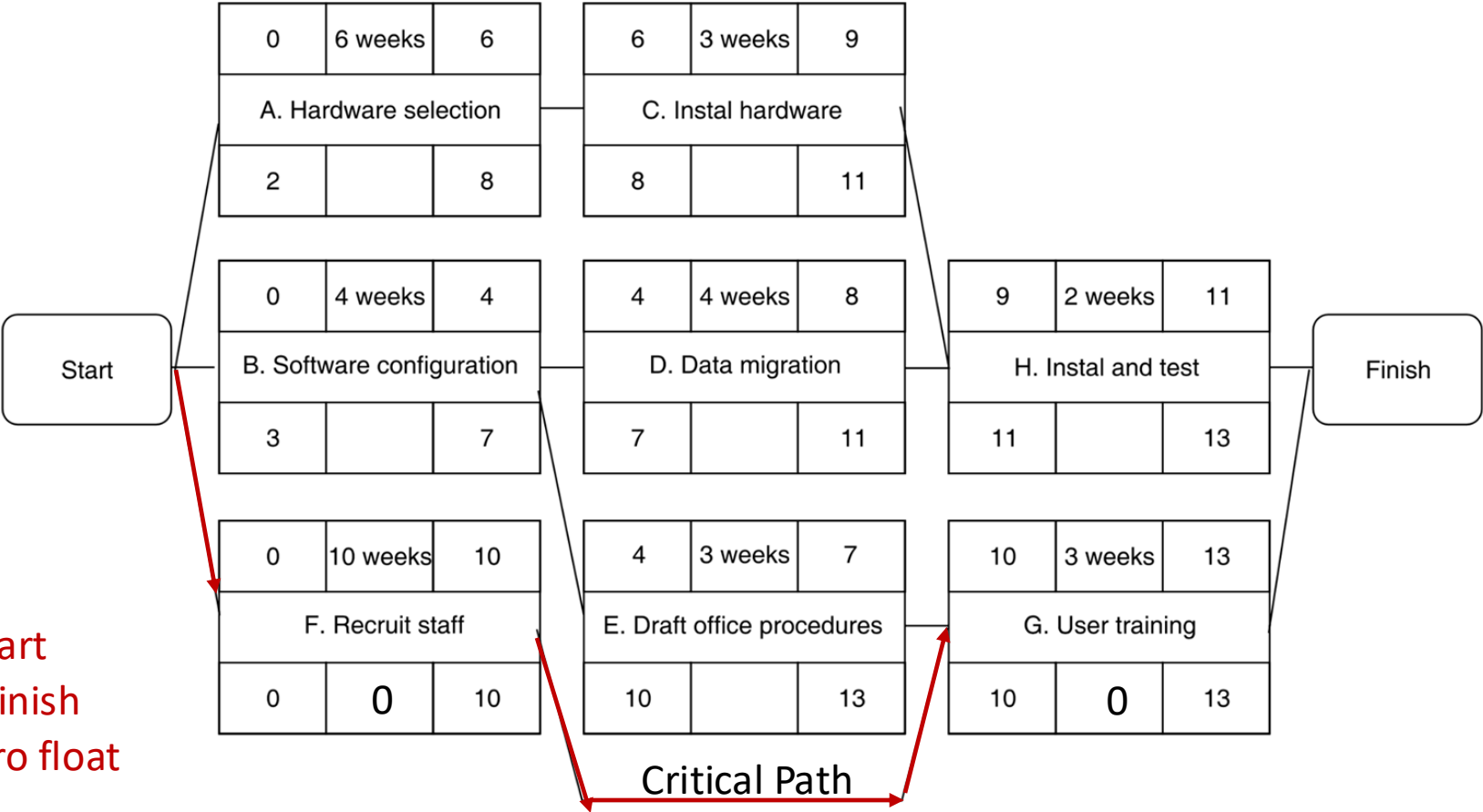
□ Critical Path

- *The critical path is the longest path through the network.*
- The difference between an activity's earliest start date and its latest start date is known as the *activity's float*.
 - *Float*: A measure of how much the start or completion of an activity may be delayed without affecting the end date of the project. *Any activity with a float of zero is critical in the sense that any delay in carrying out the activity will delay the completion date of the project as a whole.*
 - **Total Float**: The amount of time that a schedule activity can be delayed or extended from its early start date without delaying the project finish date or violating a schedule constraint.
 - **Total Float or Float = $LS - ES$ or $LF - EF$**
 - **Free Float**: The amount of time that a schedule activity can be delayed without delaying the early start date of any successor or violating a schedule constraint
 - **Free Float = $ES \text{ (of successor)} - EF \text{ (of current)}$**

Identifying Critical Path:

Critical path is the path which gives us or helps us to estimate the earliest time in which the whole project can be completed. Any delay to an activity on this critical path will lead to a delay in the completion of the whole project. In order to identify the critical path, we need to calculate the activity float for each activity.

Float = Latest Start - Earliest Start
Float = Latest Finish - Earliest Finish
Critical path = Path that has zero float



Activity float is actually the difference between an activity's Earliest start and its latest start date or the difference between the activity's Earliest finish and its latest finish date and it indicates that how much the activity can be delayed without delaying the completion of the whole project. If the float of an activity is zero, then the activity is an critical activity and must be added to the critical path of the project network. In this example, activity F and G have zero float and hence, are critical activities.

Earliest start	Duration	Earliest finish
Activity label, activity description		
Latest start	Float	Latest finish

Network Planning Model (CPM)

□ Example 2: Find the critical path

Task ID	Task	Dependency	Estimate (Days)
1	Collect contacts for publications	-	0.5
2	Create invite	-	0.5
3	Send out invites	1, 2	0.5
4	Confirm speakers and panelists	-	1
5	Check speaker slides	4	0.5
6	Press release outline	-	1
7	Collect assets	-	1
8	Draft press release	6	1
9	Finalize press release	8	3
10	Send out press kit	9, 1, 7	1
11	Implement press conference	10, 5, 3	0.5

Network Planning Model (PERT)

- PERT: Program Evaluation and Review Technique
- Take account of the uncertainty surrounding estimates of task durations.
- The method is very similar to the CPM technique but, instead of using a single estimate for the duration of each task, PERT requires three estimates:
 - **Most likely time**: the time we would expect the task to take under normal circumstances. We shall identify this by the letter *m*.
 - **Optimistic time**: the shortest time in which we could expect to complete the activity. We shall use the letter *a* for this.
 - **Pessimistic time**: the worst possible time, allowing for all reasonable eventualities but excluding 'acts of God and warfare' (as they say in most insurance exclusion clauses). We shall call this *b*.
- PERT then combines these three estimates to form a single expected duration, t_e , using the formula

$$t_e = (a + 4m + b) / 6$$

Network Planning Model (PERT)

- The expected durations are used to carry out a forward pass through a network, using the same method as the CPM technique. *In this case, however, the calculated event dates are not the earliest possible dates but the dates by which we expect to achieve those events.*

Network Planning Model (PERT)

□ Example

Activity	Optimistic (<i>a</i>)	Activity durations (weeks). Most likely (<i>m</i>)	Pessimistic (<i>b</i>)
A	5	6	8
B	3	4	5
C	2	3	3
D	3.5	4	5
E	1	3	4
F	8	10	15
G	2	3	4
H	2	2	2.5

Network Planning Model (PERT)

Activity	Activity durations (weeks)			
	Optimistic (a)	Most likely (m)	Pessimistic (b)	Expected (t_e)
A	5	6	8	6.17
B	3	4	5	4.00
C	2	3	3	2.83
D	3.5	4	5	4.08
E	1	3	4	2.83
F	8	10	15	10.50
G	2	3	4	3.00
H	2	2	2.5	2.08

Network Planning Model (PERT)

Activity	Description	Predecessors	Optimistic Duration (To)	Pessimistic Duration (Tp)	Most likely Duration (Tm)
0	Start Milestone	-	0	0	0
A	Select Technical Staff	0	12	18	15
B	Site Survey	0	6	12	9
C	Select Equipments	A	9	15	12
D	Prepare Designs	B	6	18	9
E	Bring Utilities to the Site.	B	18	36	30
F	Interview Applicants and Fill Positions	A	9	15	12
G	Purchase the Equipment.	C	36	42	36
H	Construct the Power Plant	D	42	54	48
I	Develop an Information System.	A	6	18	12
J	Install the Equipment.	H,G,E	3	9	6
K	Train the Staff to Run the System	F,J,I	3	9	6

Network Planning Model (PERT)

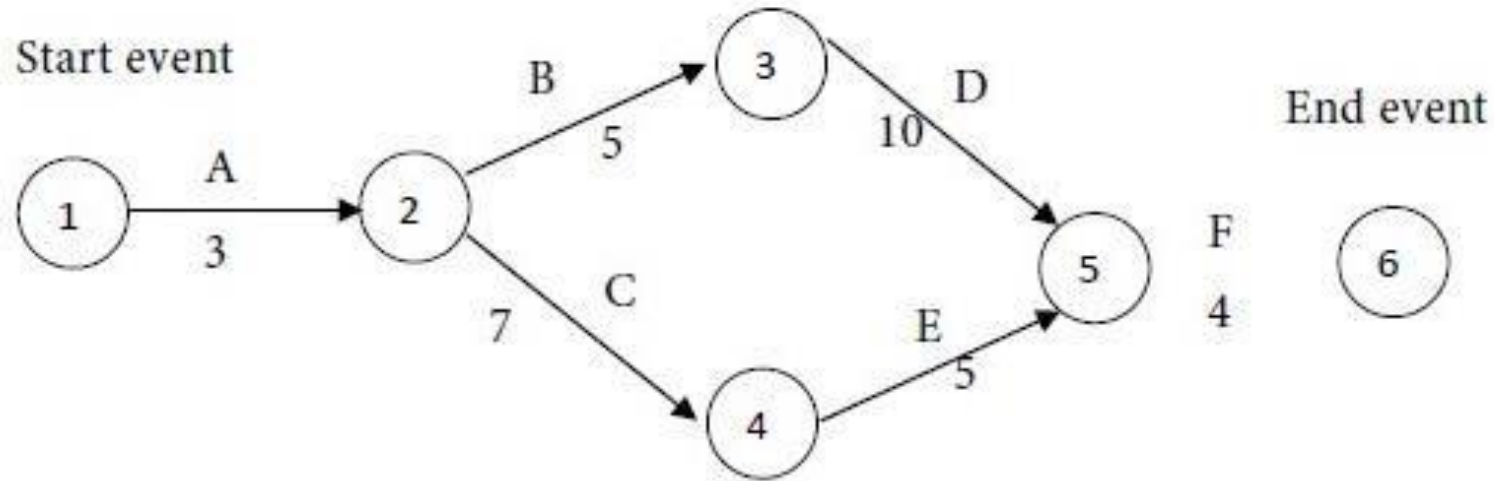
- Determine critical path
- Compute variance and standard deviation
 - *Need to compute the variance (and std. dev) for activities in Critical Path only.*
- Calculate probability of completing the project in 82 days.

Critical Path : The critical path refers to the **longest stretch** of the activities, and a measure of them from start to finish.

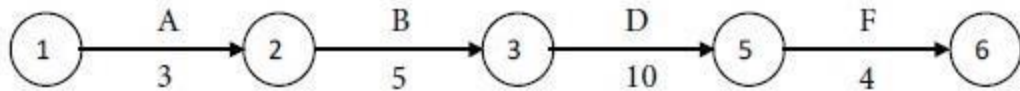
The following details are available regarding a project:

Activity	Predecessor Activity	Duration (Weeks)
A	-	3
B	A	5
C	A	7
D	B	10
E	C	5
F	D,E	4

Cont.

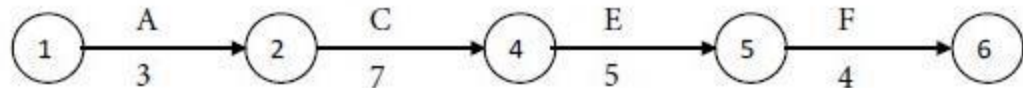


Path I



$$3+5+10+4=22 \text{ weeks}$$

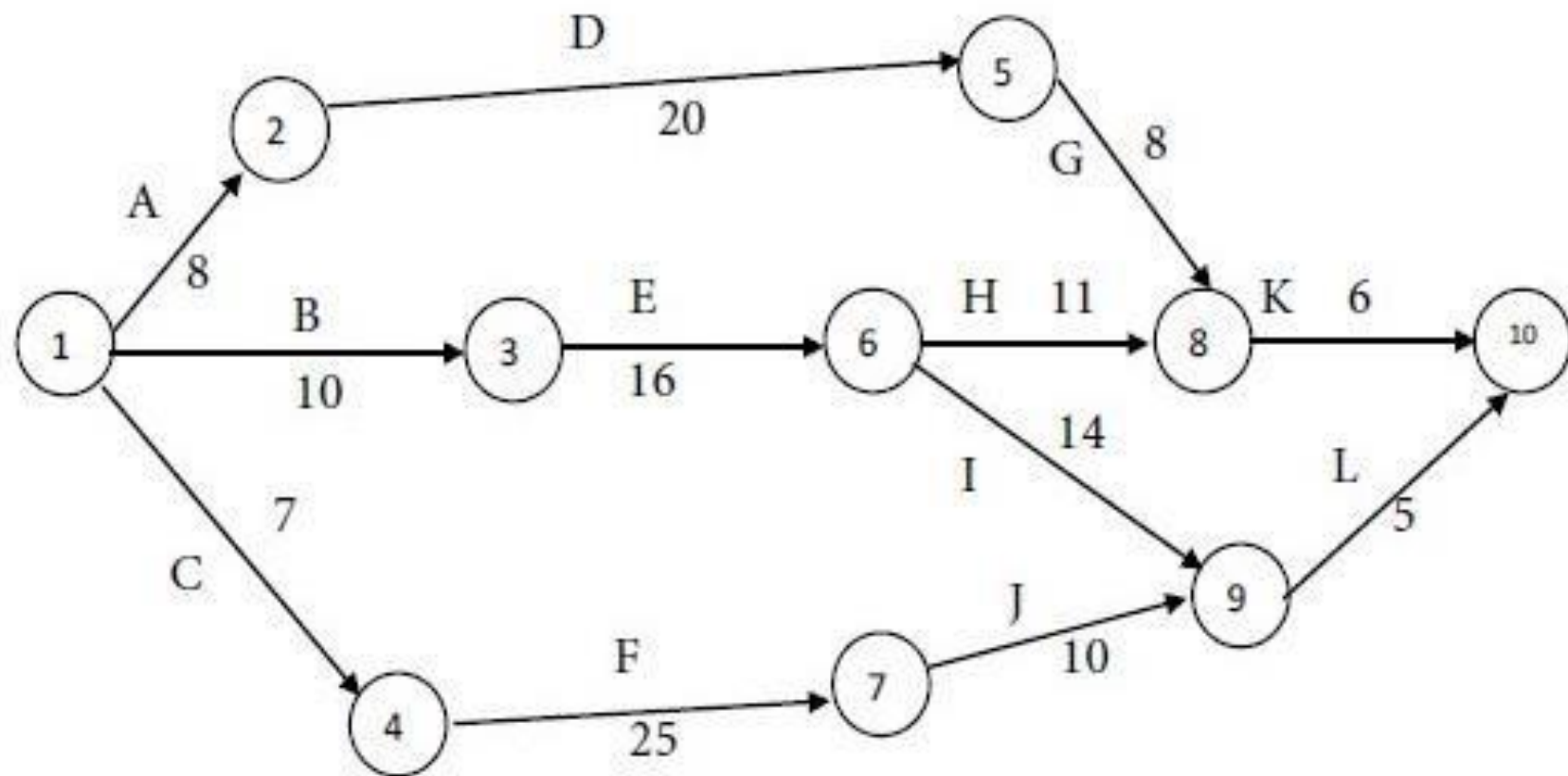
Path II



$$3+7+5+4=19 \text{ weeks}$$

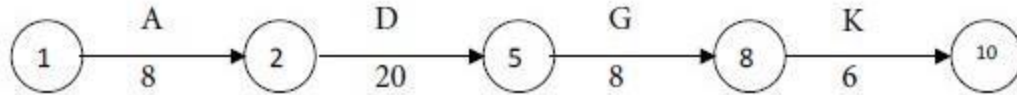
Hence critical Path is Path I with 22 Weeks.

Find out the completion time and the critical activities for the following project:



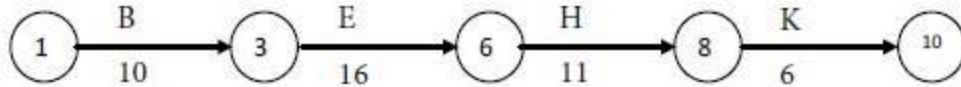
Solution

Path I



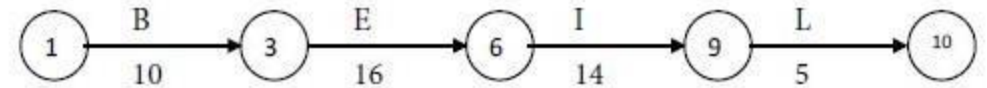
Time for the path = $8 + 20 + 8 + 6 = 42$ units of time.

Path II



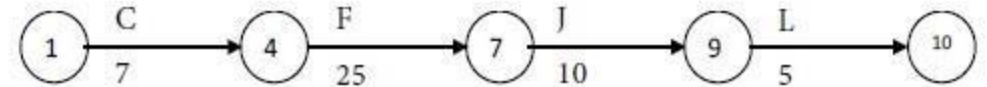
Time for the path = $10 + 16 + 11 + 6 = 43$ units of time.

Path III



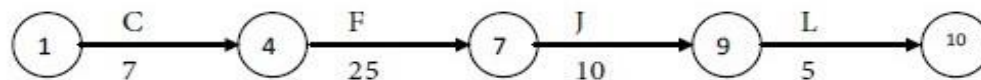
Time for the path = $10 + 16 + 14 + 5 = 45$ units of time.

Path IV



Time for the path = $7 + 25 + 10 + 5 = 47$ units of time.

Compare the times for the four paths. Maximum of $\{42, 43, 45, 47\} = 47$. We see that the following path has the maximum time and so it is the critical path:



The critical activities are C, F, J and L. The non-critical activities are A, B, D, E, G, H, I and K. The project completion time is 47 units of time.

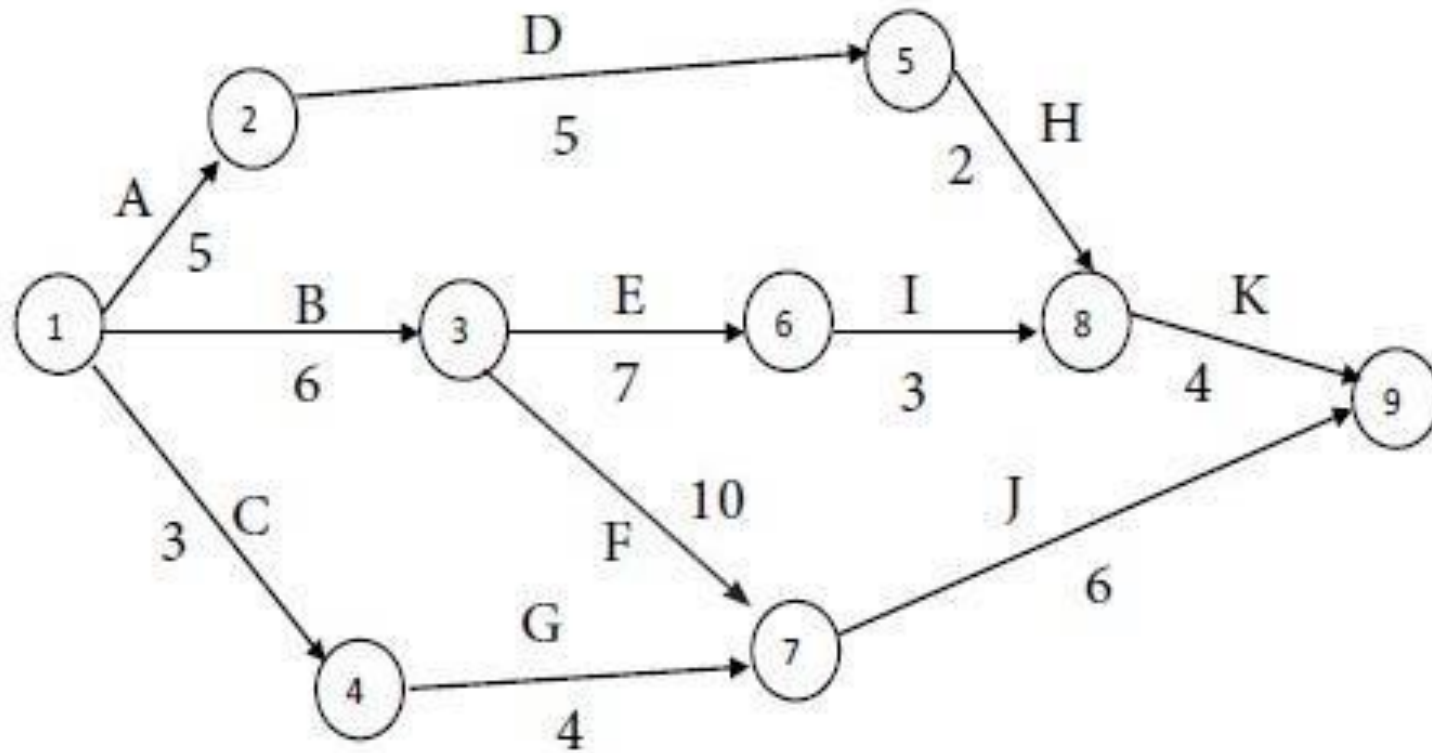
Cont.

Draw the network diagram and determine the critical path for the following project:

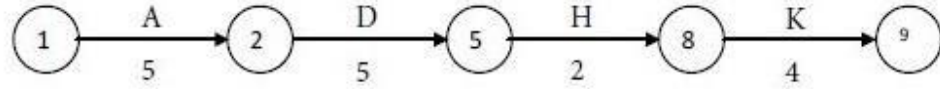
Activity	Time estimate (Weeks)
1 - 2	5
1 - 3	6
1 - 4	3
2 - 5	5
3 - 6	7
3 - 7	10
4 - 7	4
5 - 8	2
6 - 8	5
7 - 9	6
8 - 9	4

Question and Solution

We have the following network diagram for the project:

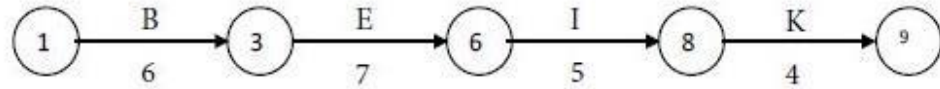


Path I



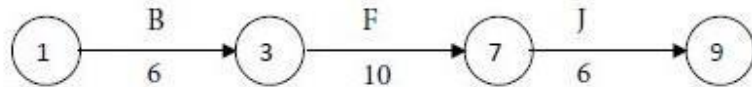
Time for the path = $5 + 5 + 2 + 4 = 16$ weeks.

Path II



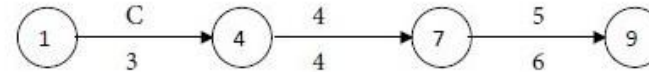
Time for the path = $6 + 7 + 5 + 4 = 22$ weeks.

Path III



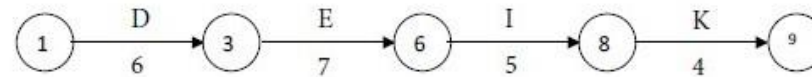
Time for the path = $6 + 10 + 6 = 16$ weeks.

Path IV



Time for the path = $3 + 4 + 6 = 13$ weeks.

Compare the times for the four paths. Maximum of $\{16, 22, 16, 13\} = 22$. We see that the following path has the maximum time and so it is the critical path:



The critical activities are B, E, I and K. The non-critical activities are A, C, D, F, G, H and J. The project completion time is 22 weeks.

PERT Networks (cont'd)

- Formula for Calculating Estimated PERT Times

$$T_e = \frac{T_o + 4T_m + T_p}{6}$$

T_e = Time estimate (average)

T_o = Time estimate (optimistic)

T_m = Time estimate (most likely)

T_p = Time estimate (pessimistic)

Standard Deviation of an Activity, $\sigma = \frac{P - O}{6}$

Variance of an Activity, $\sigma^2 = \left(\frac{P - O}{6}\right)^2$

Condition in Critical Path

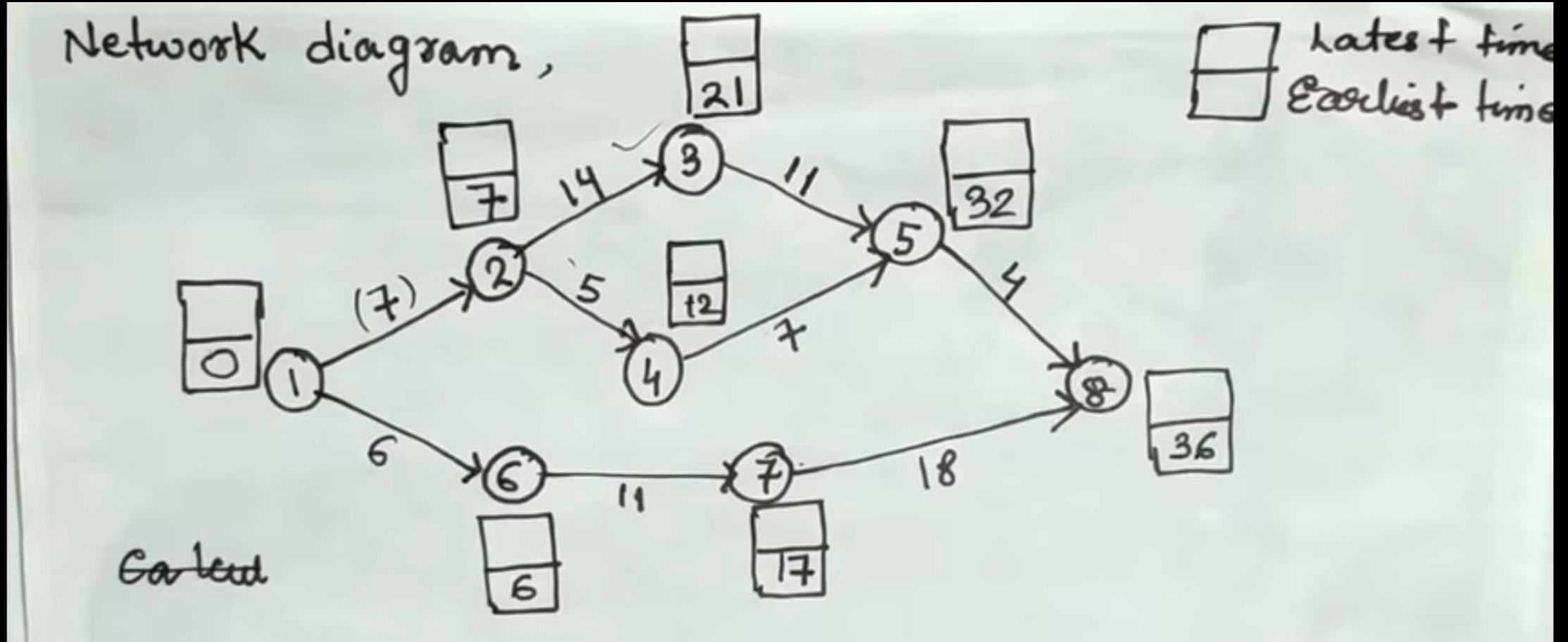
$$\left. \begin{array}{l} E_i = L_i \\ E_j = L_j \end{array} \right\} \\ E_j - E_i = L_j - L_i = D_{ij}$$

PERT (Program Evaluation and Review Technique)

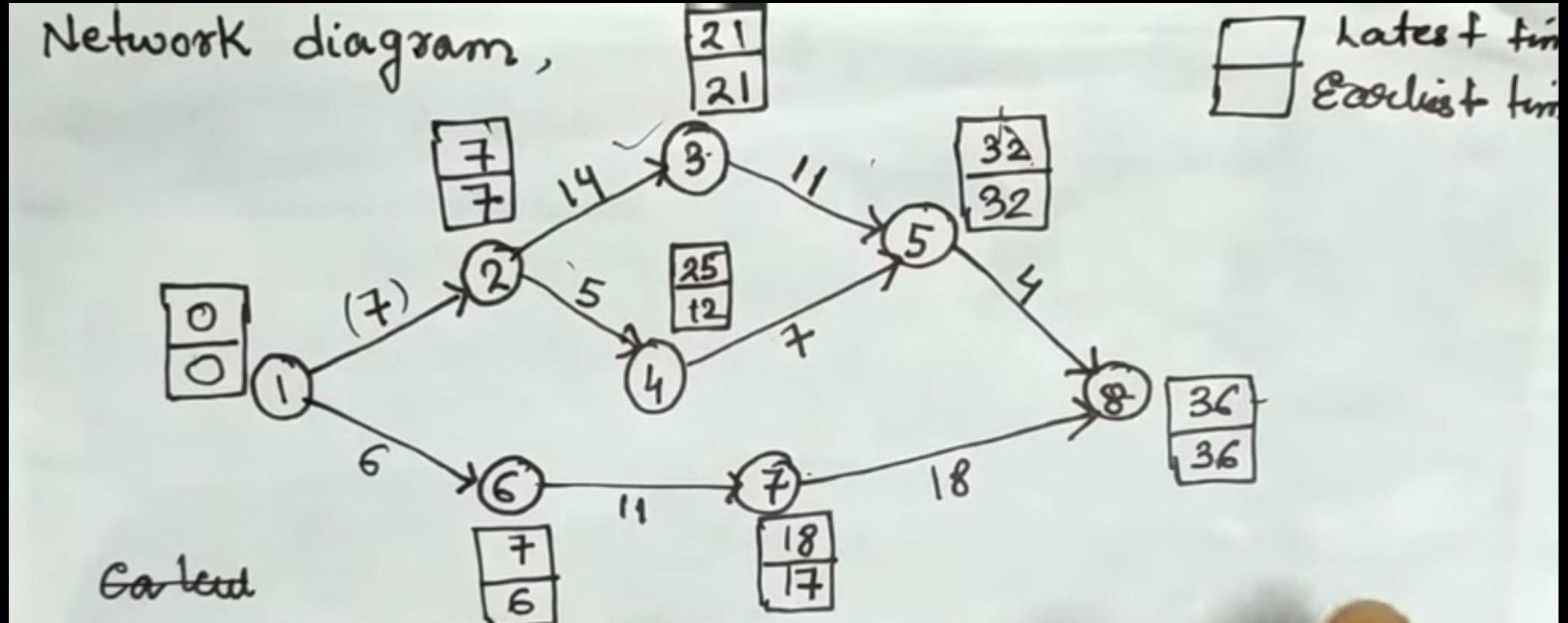
Activity	Estimated Duration (Weeks)		
	Optimistic time (To)	Most Likely Time (Tm)	Pessimistic time (Tp)
1-2	1	7	13
1-6	2	5	14
2-3	2	14	26
2-4	2	5	8
3-5	7	10	19
4-5	5	5	17
6-7	5	8	29
5-8	3	3	9
7-8	8	17	32

Activity	Estimated Duration (Weeks)			$T_e = \frac{T_0 + 4T_m + T_p}{6}$	$\sigma^2 = \left(\frac{T_P - T_0}{6}\right)^2$
	Optimistic time (To)	Most Likely Time (Tm)	Pessimistic time (Tp)		
1-2	1	7	13	7	4
1-6	2	5	14	6	4
2-3	2	14	26	14	16
2-4	2	5	8	5	1
3-5	7	10	19	11	4
4-5	5	5	17	7	4
6-7	5	8	29	11	16
5-8	3	3	9	8	1
7-8	8	17	32	18	16

Earliest Time



Latest Time



Critical Path=1-2-3-5-8

Expected project duration= $7+14+11+4=36$ weeks

Project length variance= $4+16+4+1=25$

Standard Deviation=5